

CSCC43: Introduction to Databases

Project Written Tasks - Fall 2026

These four written tasks are the steps to design a database you learn from this course. They assess how well you have learned database design skills and also prepare you for implementing the social network and portfolio management application in the project. You are supposed to work on these tasks together with your project teammate and submit a report (in PDF format up to 8 pages) to Quercus.

Task 1 - E/R Modeling

Draw the E/R diagram for the application. The diagram in the submission must be generated by a drawing software tool, e.g., draw.io or Microsoft PowerPoint. Specify any assumptions you make.

Task 2 - Relational Schema

Develop the relational schema for the database based on your E/R diagram.

Task 3 - Normalization

Optimize your database with normalization. Show that each table in the database is in BCNF.

Task 4 - Queries in Relational Algebra

Pick five queries that you will use to implement the core application logic in relational algebra. For each query, specify the logic and show the relational algebra expression(s).